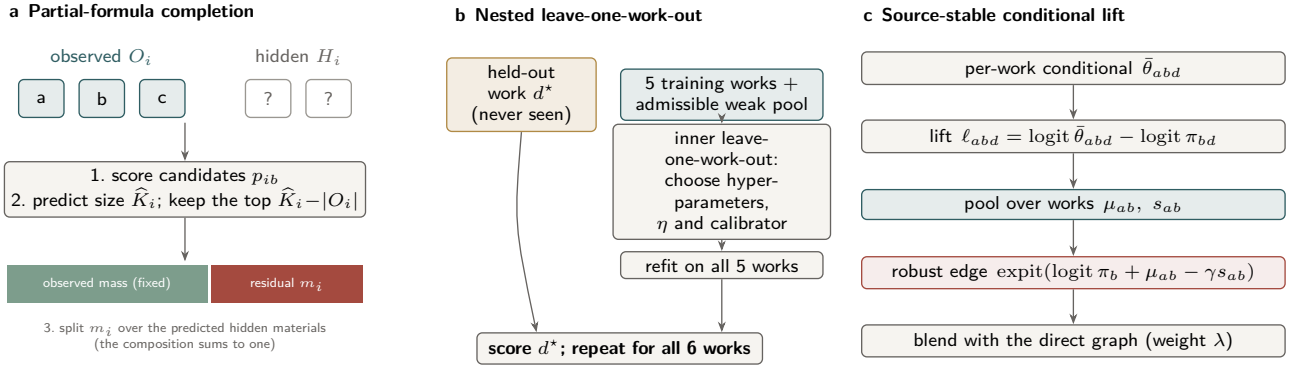
a Partial-formula completion
observed $O_i$
hidden $H_i$
a
b
c
?
?
1. score candidates $p_{ib}$
2. predict size $\widehat{K}_i$; keep the top $\widehat{K}_i - |O_i|$
observed mass (fixed)
residual $m_i$
3. split $m_i$ over the predicted hidden materials
(the composition sums to one)
b Nested leave-one-work-out
held-out
work $d^\star$
(never seen)
5 training works +
admissible weak pool
inner leave-
one-work-out:
choose hyper-
parameters,
$\eta$ and calibrator
refit on all 5 works
score $d^\star$; repeat for all 6 works
c Source-stable conditional lift
per-work conditional $\bar{\theta}_{abd}$
lift $\ell_{abd} = \operatorname{logit} \bar{\theta}_{abd} - \operatorname{logit} \pi_{bd}$
pool over works $\mu_{ab},\ s_{ab}$
robust edge $\operatorname{expit}(\operatorname{logit} \pi_b + \mu_{ab} - \gamma s_{ab})$
blend with the direct graph (weight $\lambda$)